Practical Task 1 – eCommerce Application

1. Introduction

This project involves the development of a basic eCommerce web application using Django. The system allows users to view products, add items to a cart, place orders, and submit reviews. Vendors can create stores and add products.

2. Problem Statement

The goal of this project is to design and implement a simple online shopping platform that supports both buyers and vendors. The system should demonstrate core eCommerce functionality such as product browsing, cart management, and order processing.

3. Research

Existing eCommerce platforms such as Amazon and Shopify provide similar functionality. These systems allow users to browse products, manage carts, and complete purchases. This project aims to replicate simplified versions of these features.

4. Target Users

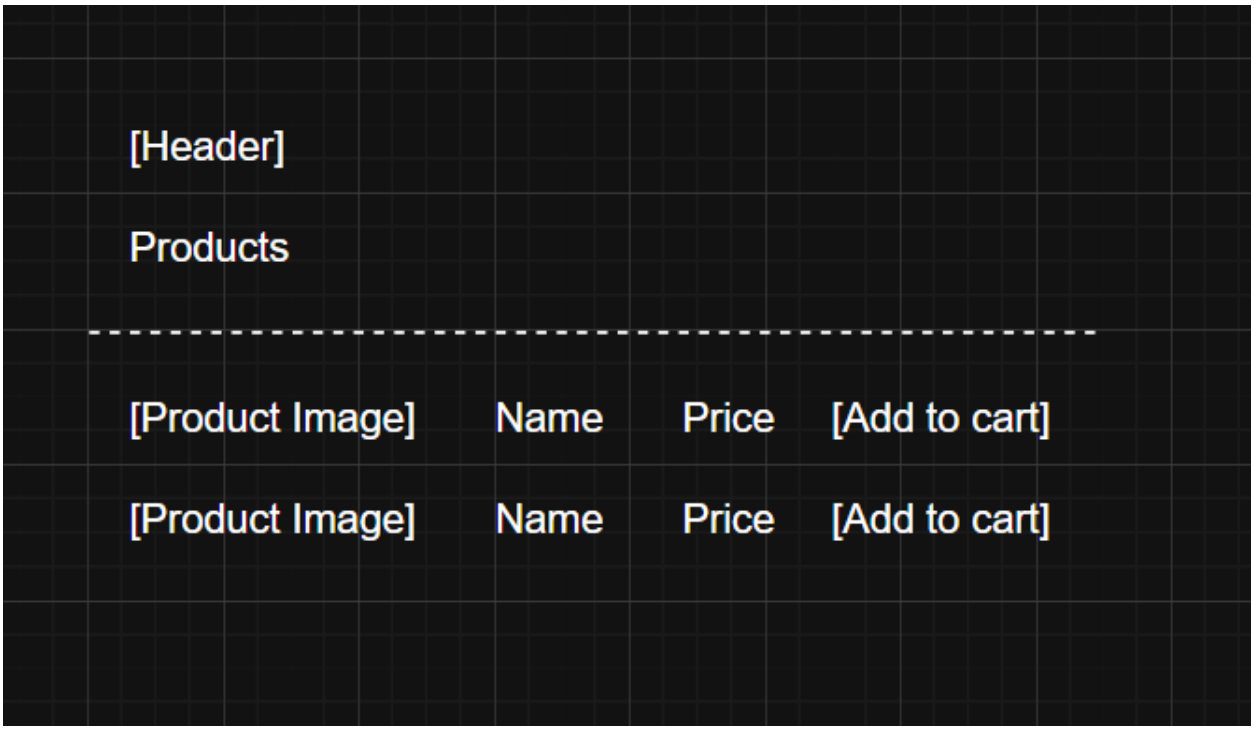
- Buyers: Users who browse products, add items to cart, and place orders
- Vendors: Users who create stores and add products

5. Features

- User registration and login
- Product listing
- Add to cart functionality
- Checkout system
- Product reviews
- Vendor dashboard (create store and add products)

6. System Design / Wireframes

6.1 Product List Page



6.2 Cart Page

Cart

Product Name - Price

Product Name - Price

[Checkout Button

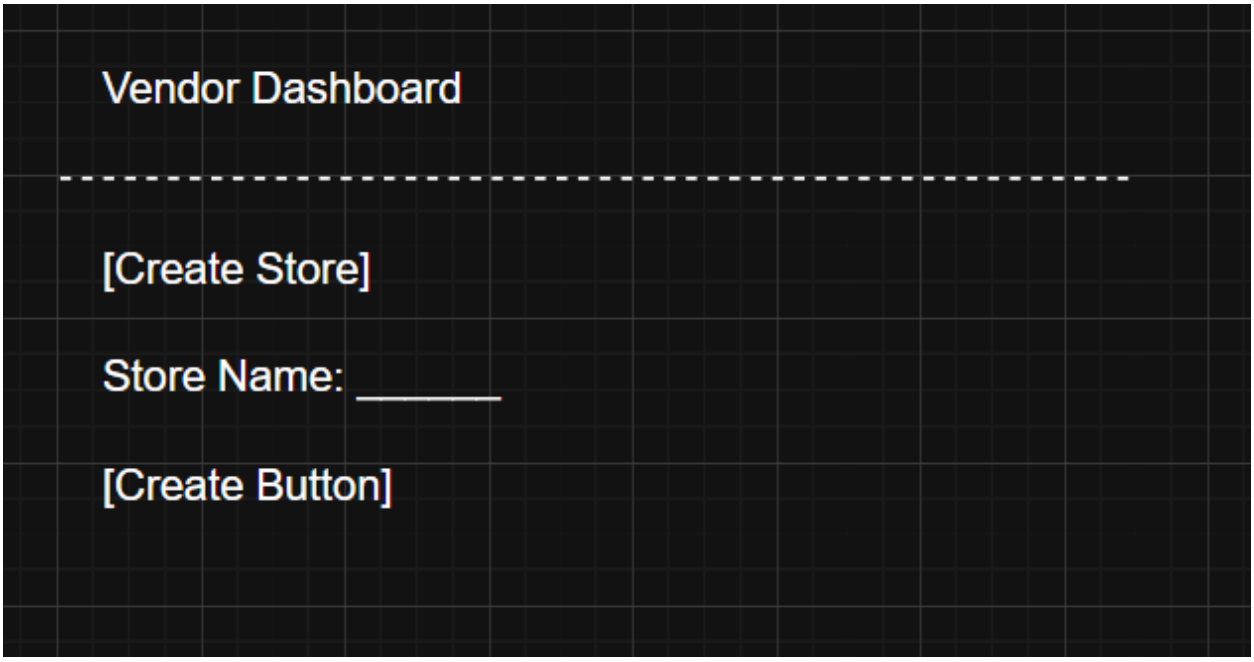
6.3 Checkout Page

Checkout

Order summary

[Confirm Order Button]

6.4 Vendor Dashboard



7. System Flow

Buyer Flow:

User → View Products → Add to Cart → Checkout → Order Created

Vendor Flow:

User → Login → Create Store → Add Product → Product Displayed

8. Implementation

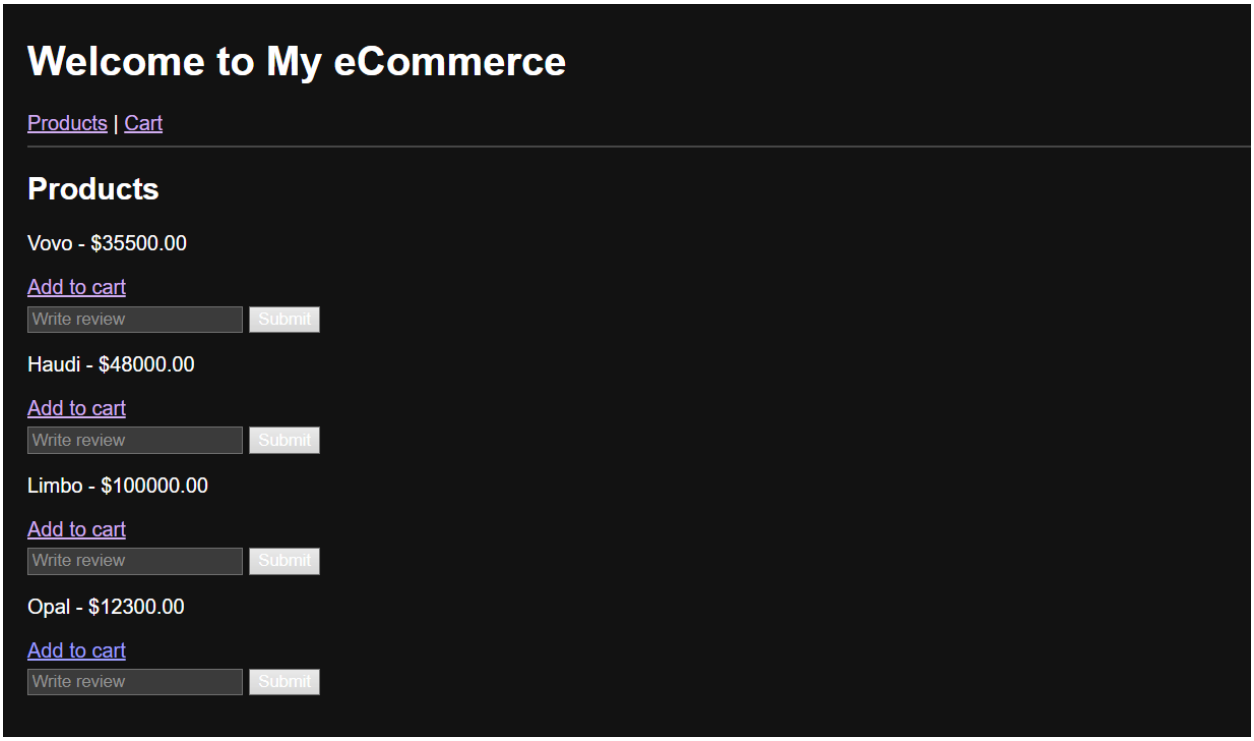
The system was built using Django. Models were created for:

- User
- Store
- Product
- Order
- Review

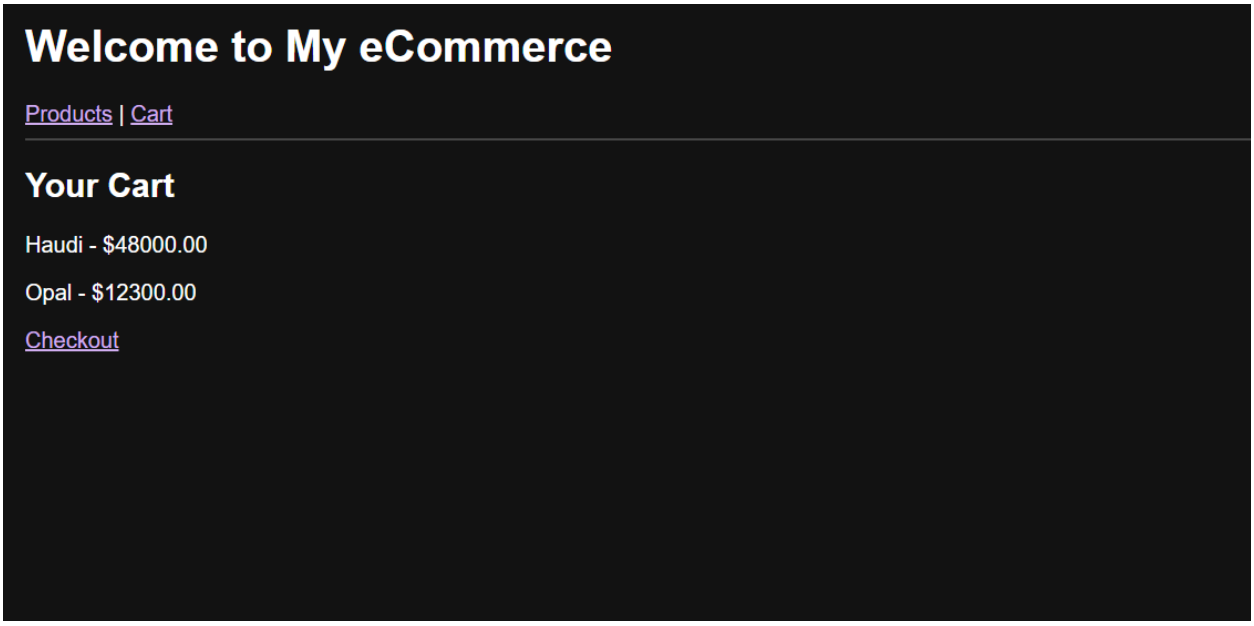
Views and templates were implemented to handle user interaction and display data.

9. Testing

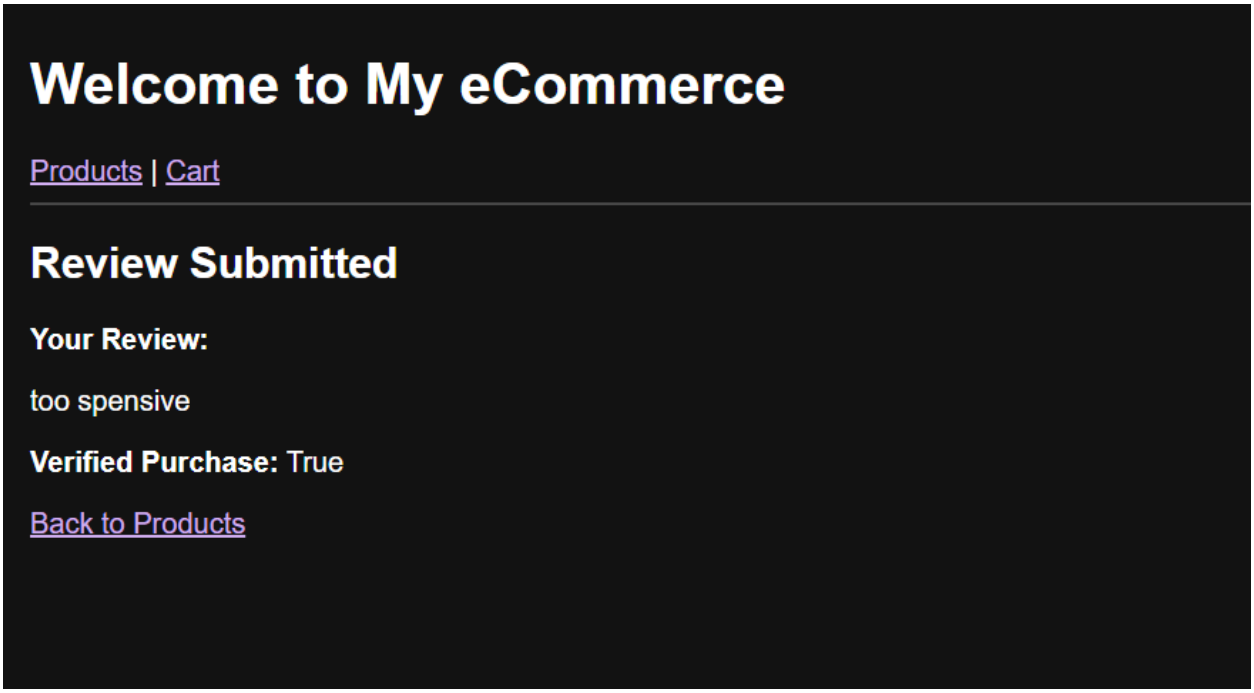
9.1 Product Page



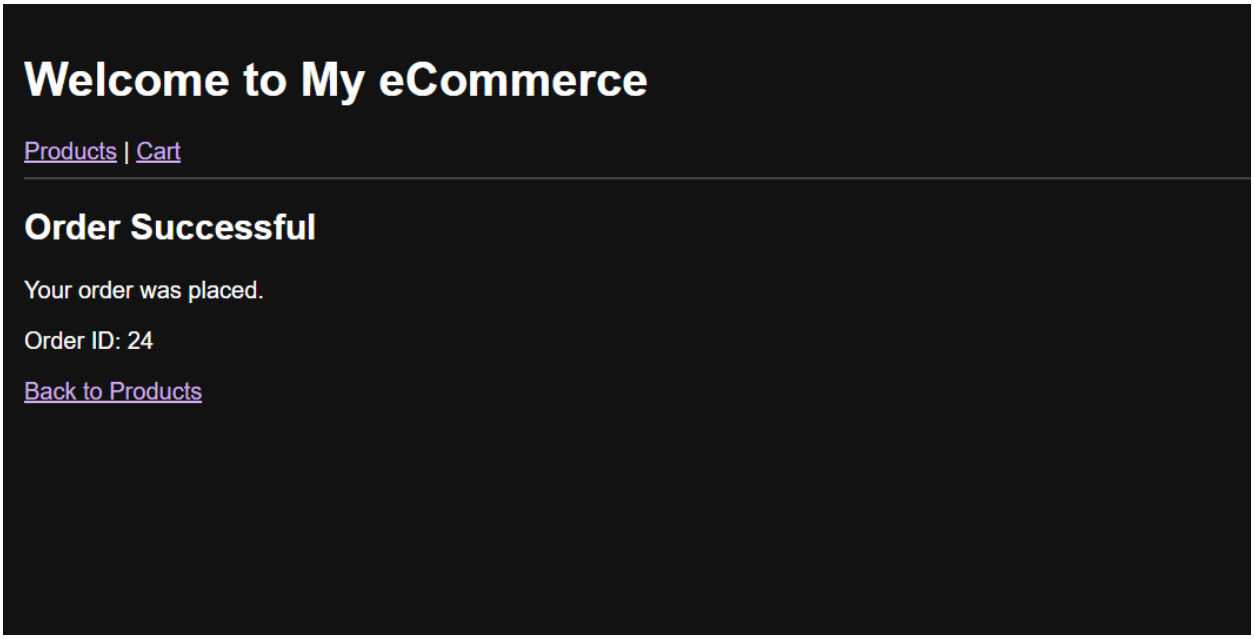
9.2 Cart Functionality



9.3 Review Submission



9.4 Checkout Process



10. Challenges

- Setting up custom user model
- Fixing URL routing issues
- Handling empty templates and missing pages

11. Reflection

This project helped in understanding Django fundamentals such as models, views, templates, and authentication. While core functionality works, improvements can be made to user interface and error handling.

12. Conclusion

The application successfully demonstrates a basic eCommerce system with essential features. Future improvements could include better UI design, payment integration, and enhanced user experience.